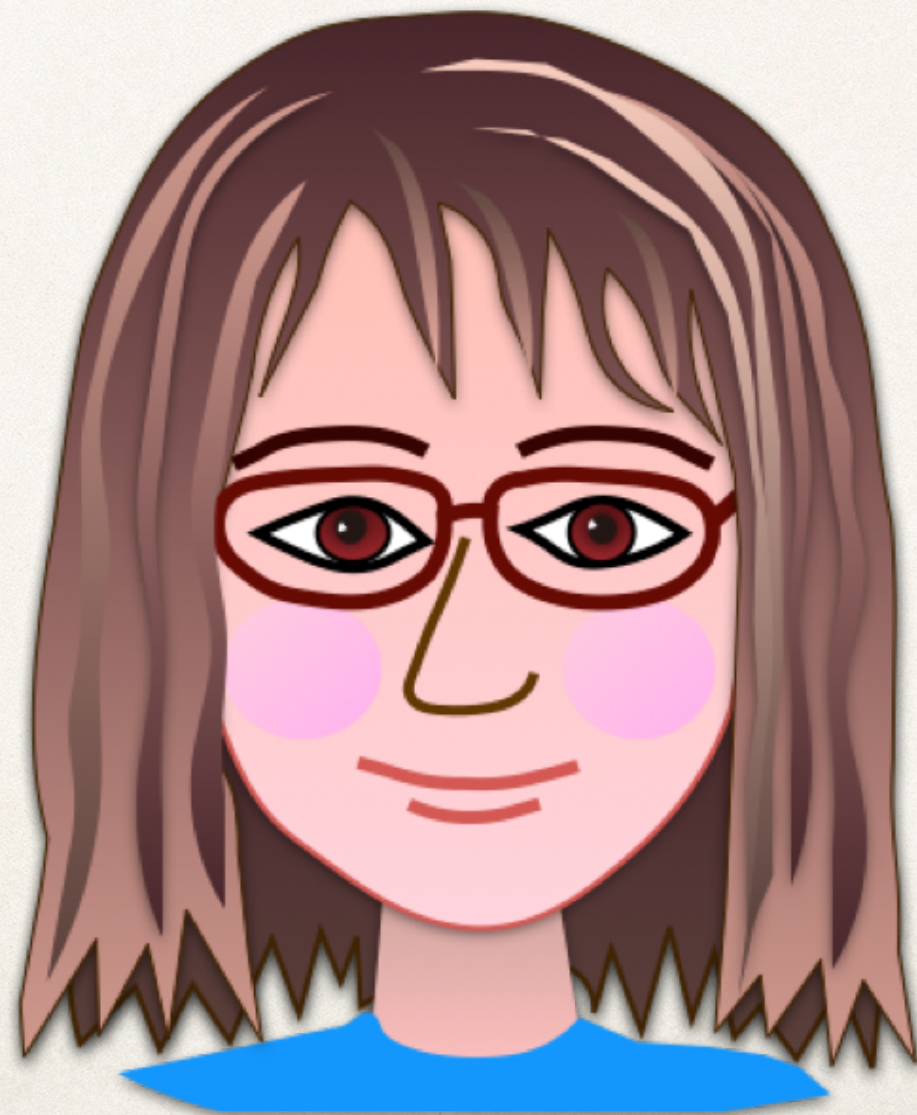
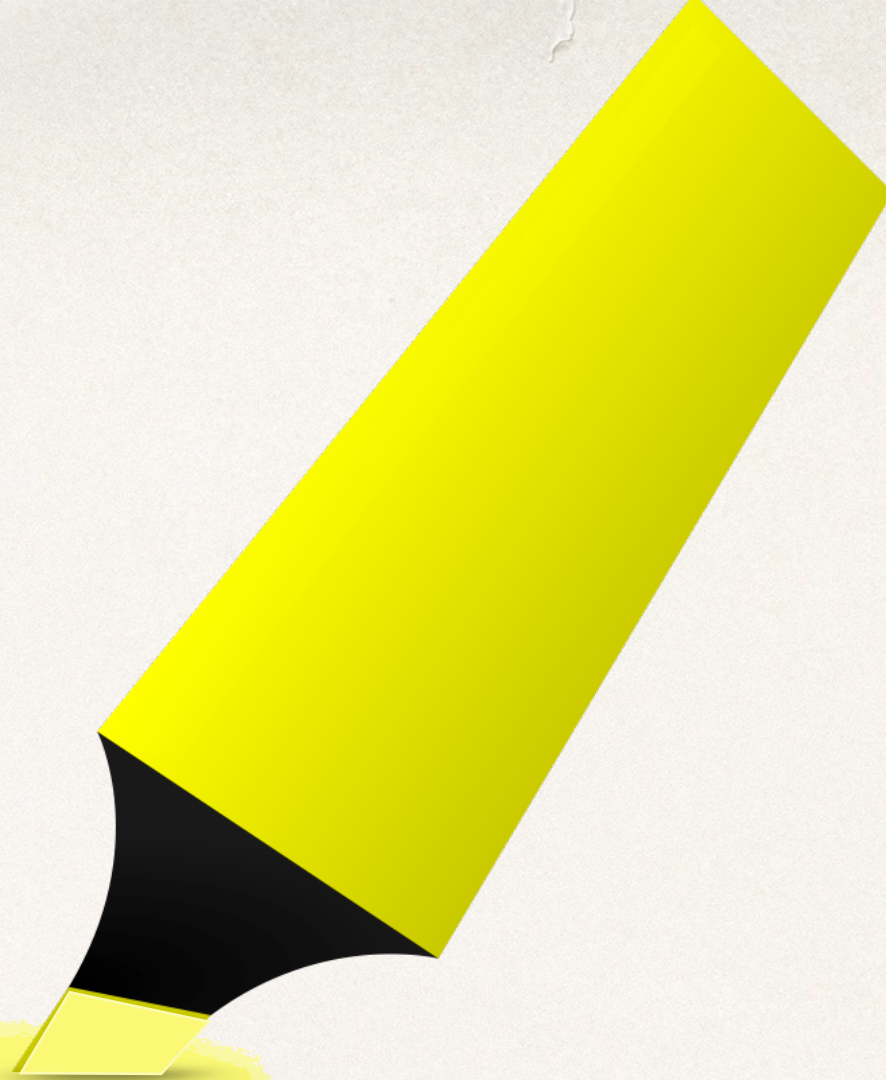
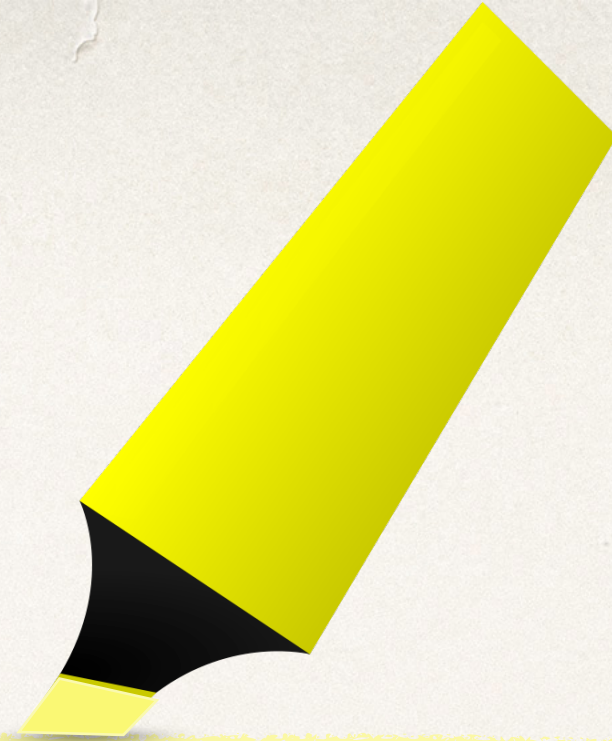


Advice for Power Learning - Based on the Principles of Psychology

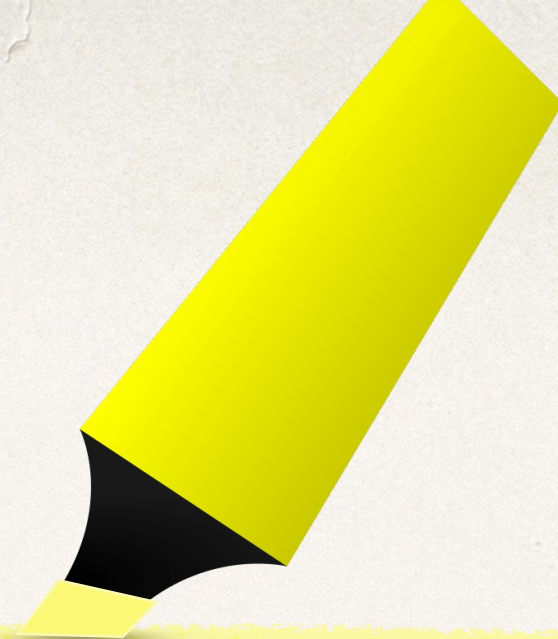




Attention is the first step
required in creating a new
memory.



**Attention refers to
concentrating on some
stimuli in your environment
while tuning others out.**



**Attention is like a
highlighter that allows you
to focus on vital information
from your environment and
ignore the rest.**

If you do not focus your attention on material, you cannot expect it to reach your long-term memory.

In fact, it will not even enter your short-term memory.

lack of system

passivity

**The rest of this presentation
will provide advice on how
to become an attentive,
active and systematic
learner.**

Attention → Short-Term Memory

**Listen, focus, rehearse, write down,
ask and answer questions.**

Add Meaning → Long-Term Memory

**Be systematic, active and creative
in your note-taking. Use categories,
symbols, examples, analogies,
pictures and quizzes.**

**Long-Term Memory
is founded on**

MEANING

1 Attention



Focus attention

by reworking and
reconstructing what you
read and hear..

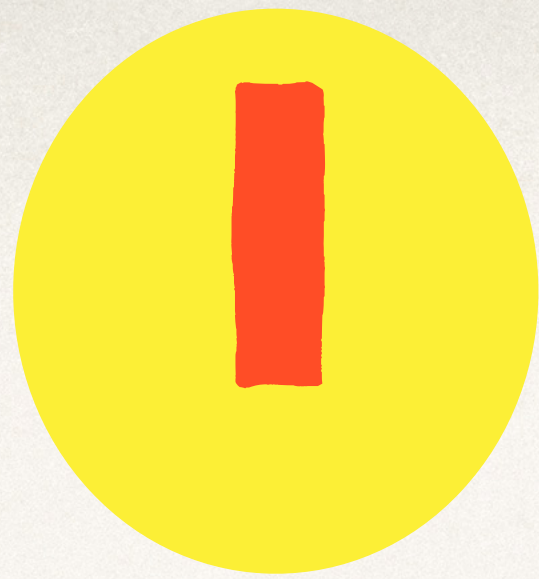
In the process, you will also

add meaning

to the new information.

Methods for

reworking and
reconstructing what you
read and hear...



Question it.
Create
questions that
you can answer
from what you
have read.

Example: You read:

**The goals of psychology
are to describe, explain,
predict and control
behaviour.**

You ask:

- When might explaining human behaviour be useful?
 - Why is predicting behaviour important?
 - What kind of behaviour might a person wish to control?
-

2

Put ideas into categories.

Psychology

Study of

- behaviour
- mental states (feelings and thoughts)

Goals

- describe
- explain
- predict
- control behaviour

Methods

- case study
- experiment
 - survey/questionnaire
- naturalistic observation
- correlational study



Use different words to reformulate an idea.

behaviour = observable action
feeling = emotion, affect
thought = cognition, belief

4

Highlight the buzz words or key terms.

behaviour

feeling

thought

= observable action

= emotion, affect

= cognition, belief

mental states

5

Think of examples,
analogies and symbols.

behaviour



kicking, playing,
laughing

feeling

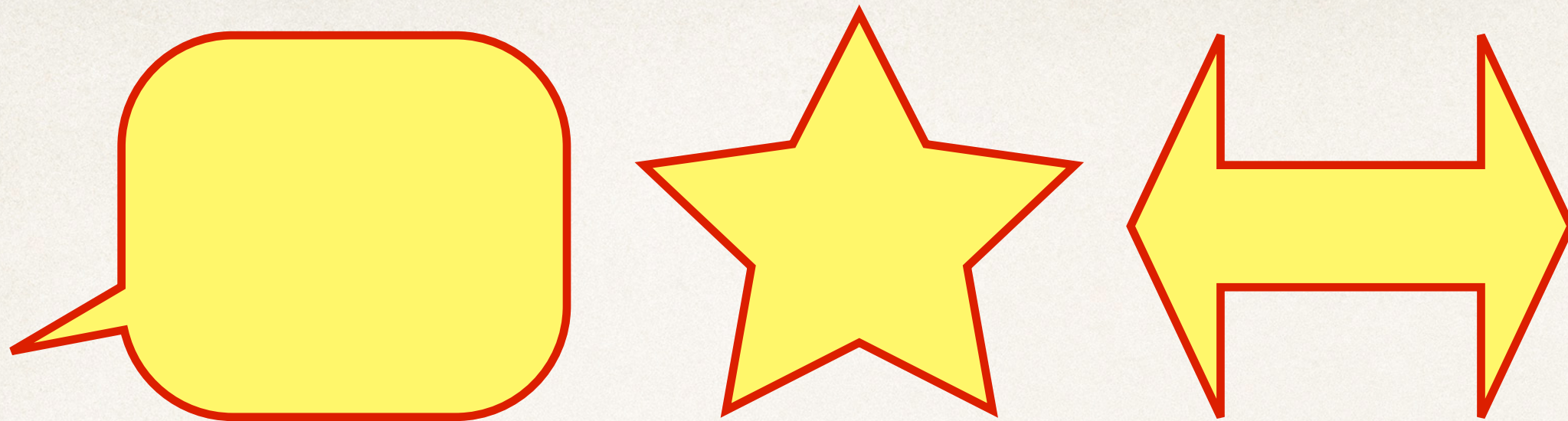


anxiety, apprehension

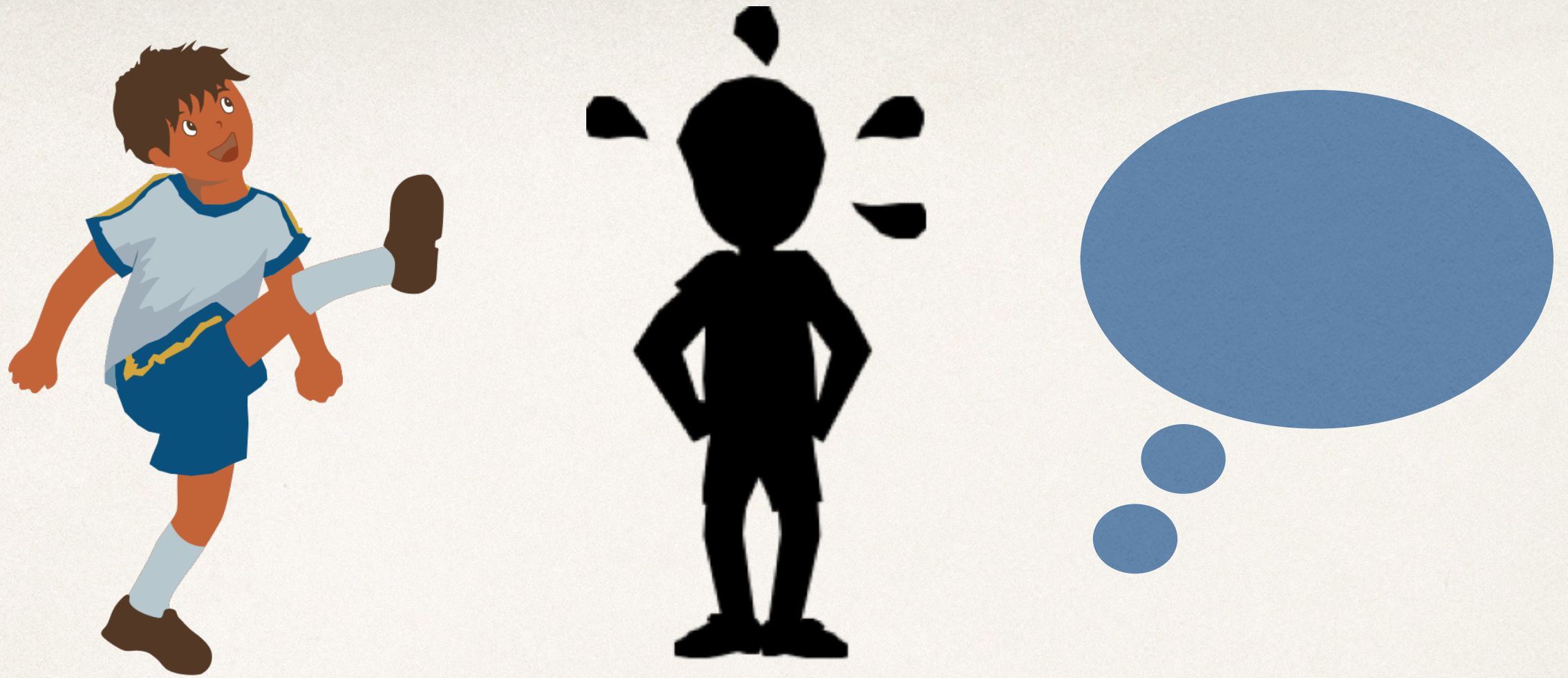
thought



idea, opinion



**Each time you think of an example,
analogy, symbol, category or
synonym, you are focusing on and
then reconstructing new
knowledge.**



This process is called ELABORATION.
It ensures that your new
knowledge is well encoded in your
memory and more easily retrieved.

Example



Psychology

Study of

- behaviour
- mental states (feelings and thoughts)

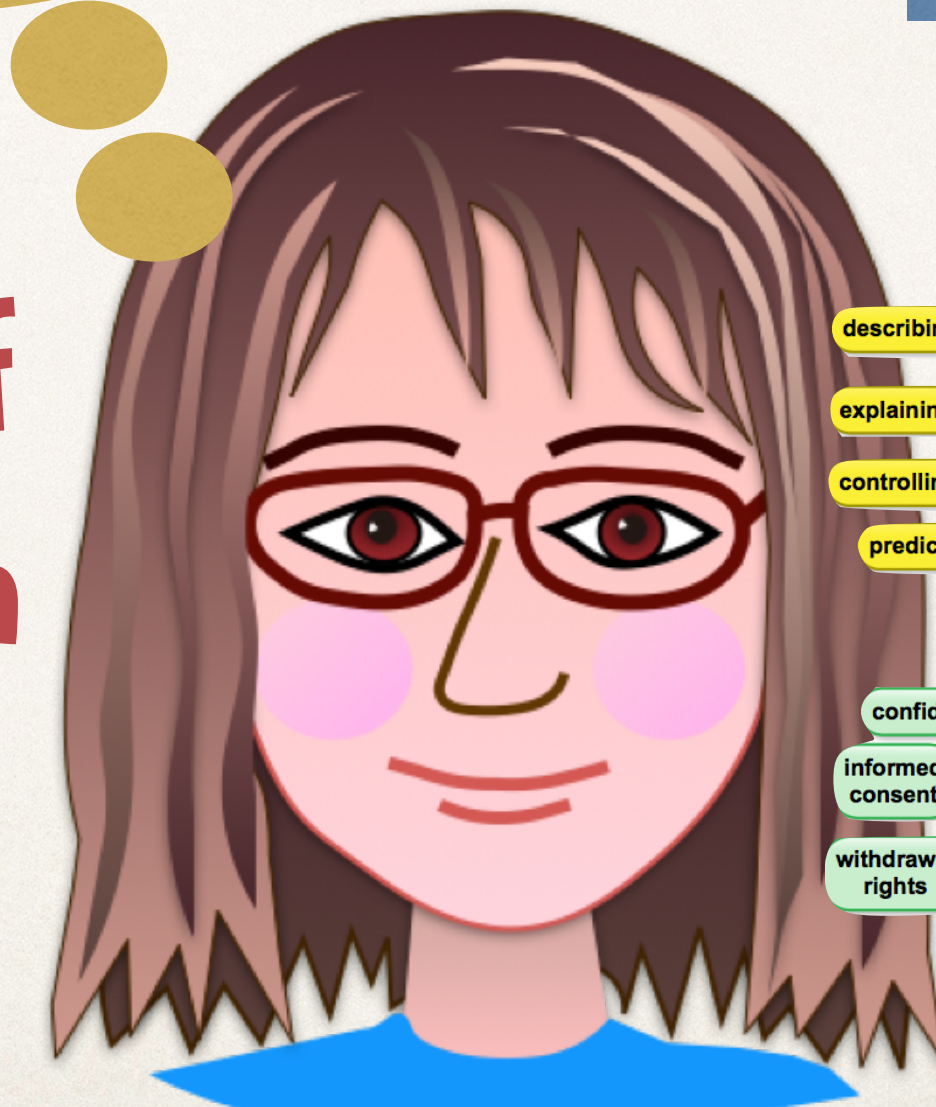
Goals

- describe
- explain
- predict
- control behaviour

Methods

- case study
- experiment
- survey/questionnaire
- naturalistic observation
- correlational study

Methods of Elaboration





6

Make yourself a CHECK QUIZ

behaviour | thought | feeling
control | describing | predict

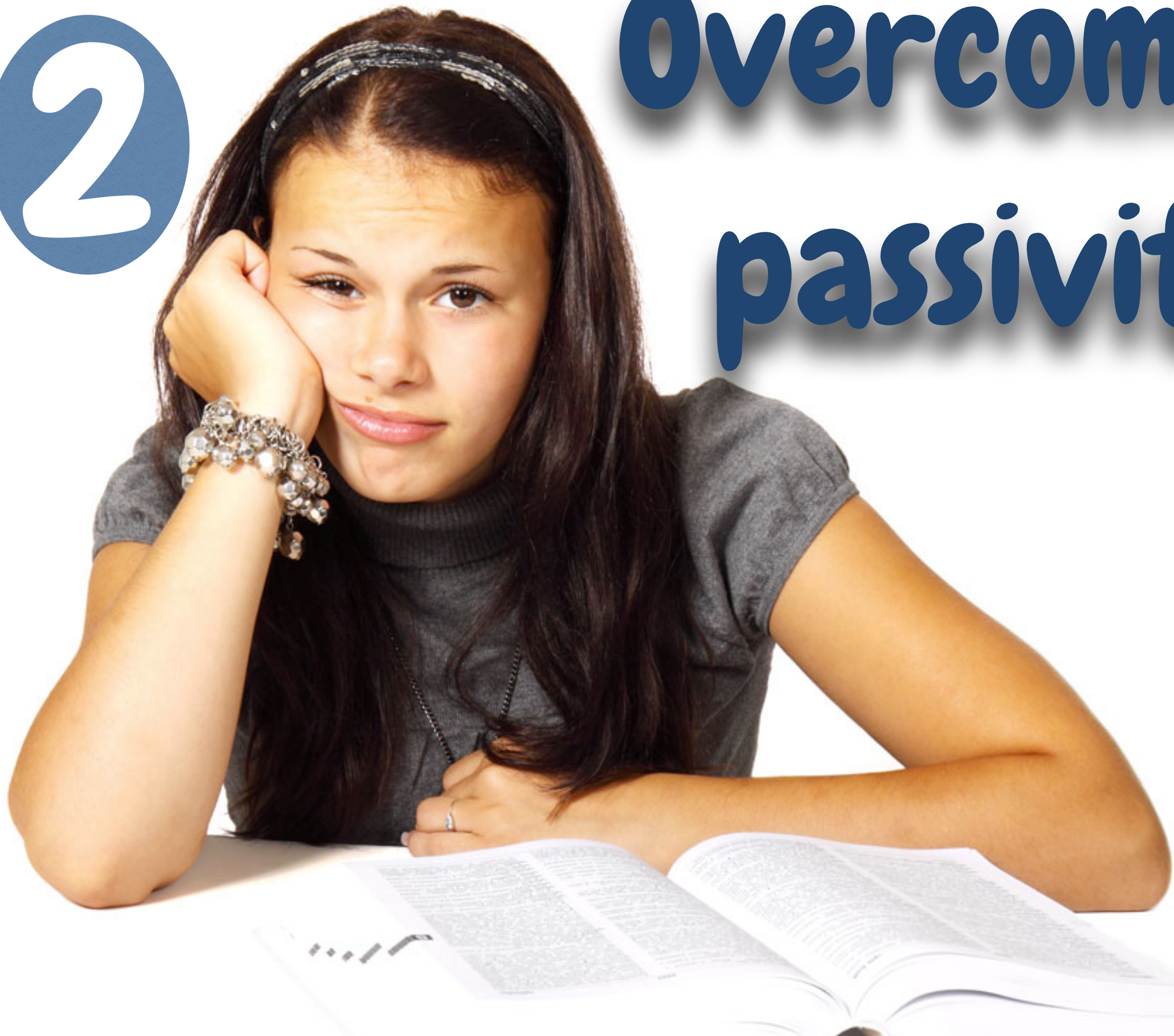
- a** A person with OCD might wish to _____
distressing behaviour.
- b** Intense fear of contamination: e.g. of _____
- c** Constant hand-washing: e.g. of _____
- d** Person's belief that certain rituals will help
protect self or family: e.g. of _____
- e** A person determining whether criminals should
be paroled might wish to _____ behaviour.
- f** Note in patient record: Person undertakes
frequent compulsive hand-washing: e.g. of _____
behaviour

ANSWERS

- a** A person with OCD might wish to control distressing behaviour.
- b** Intense fear of contamination: e.g. of feeling
- c** Constant hand-washing: e.g. of behaviour
- d** Person's belief that certain rituals will help protect self or family: e.g. of thought
- e** A person determining whether criminals should be paroled might wish to predict behaviour.
- f** Note in patient record: Person undertakes frequent compulsive hand-washing: e.g. of describing behaviour

2

Overcoming passivity



with a
SYSTEM...



Use note-taking techniques that...

1 add meaning to what you
have heard or read

For instance, think of examples to
help you understand or visualise a
new concept.

Use note-taking techniques that...

2 help you to understand
and remember the
material better



For instance, employ
symbols, diagrams, pictures,
tables and mnemonics, not
just words.

EXAMPLE: Psychology is the systematic study of...

Behaviours



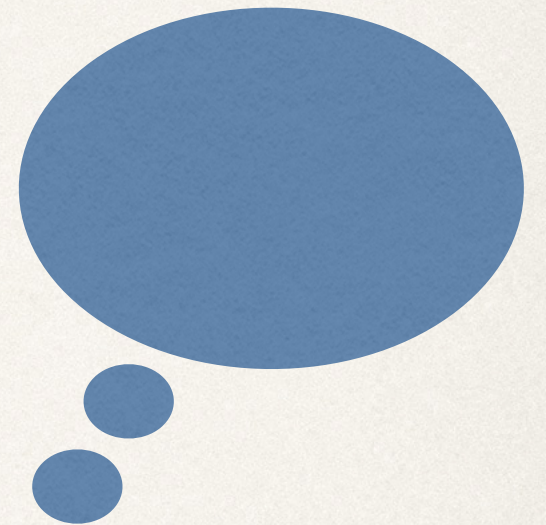
- compulsive hand-washing

Feelings



- anxiety about contamination

Thoughts

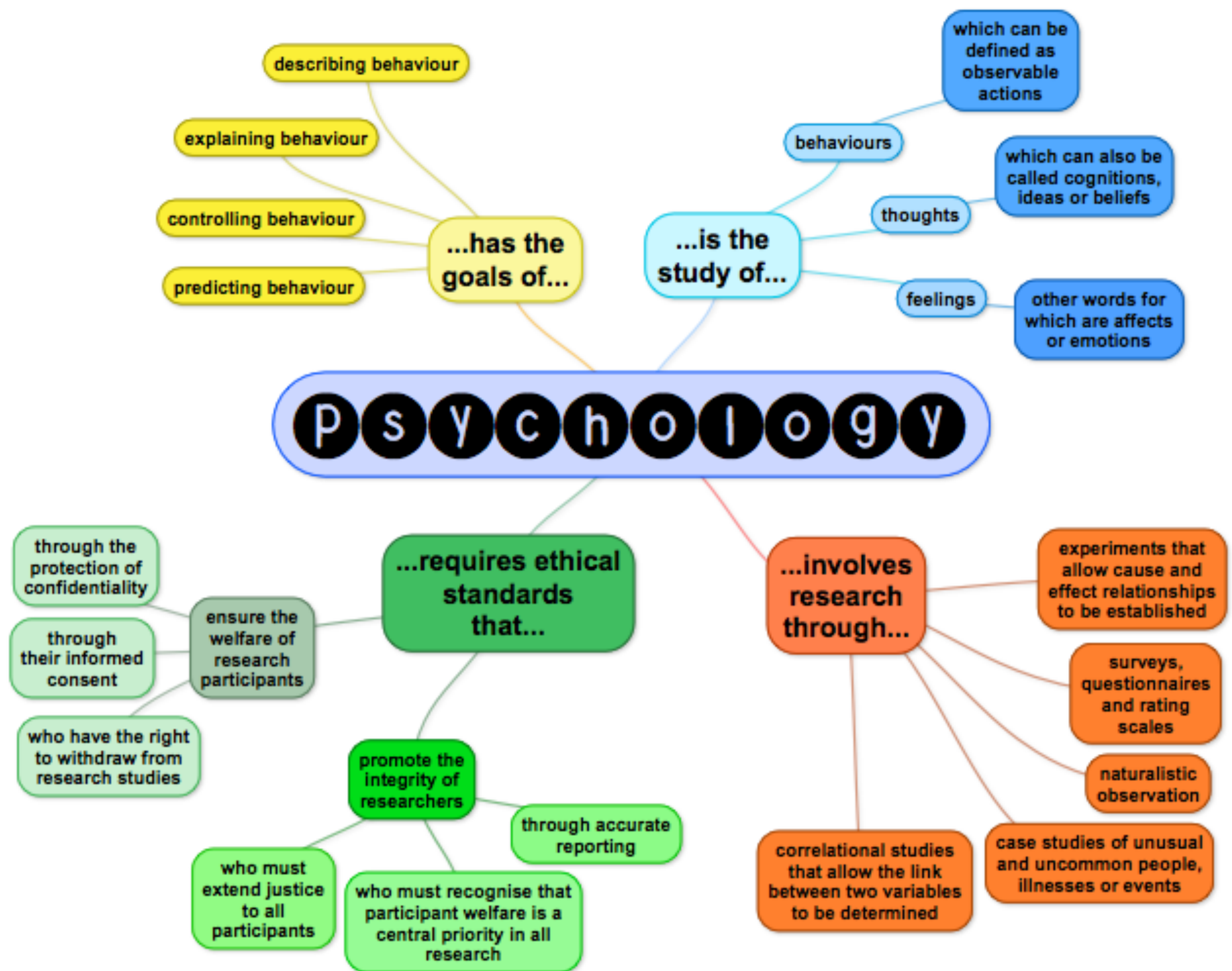


- belief that washing will protect one from contamination

Use note-taking techniques that...

3 allow you to notice and focus
on the links between ideas
and concepts

For instance, use a concept map
with links, arrows and branches.



Use note-taking techniques that...

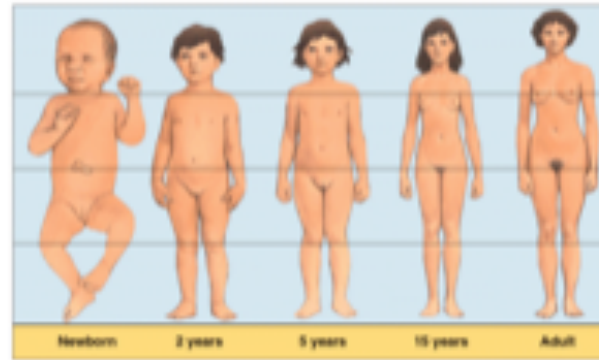
4 force you to focus on the key terms or “buzz words”

For instance, keep a running list of these words on the right of your page.

Brain Development in Infancy and Adolescence

1 Timeline of brain development

- Birth: about $\frac{1}{4}$ of adult size; almost all **neurons** already present
- 6 months: about $\frac{1}{2}$ of adult size
- 2 years: about $\frac{3}{4}$ of adult size
- 6 years: 90-95% of adult size
- Mid-20s: full adult size but some parts still maturing



The genetically programmed development of the brain, which occurs in a predetermined sequence in all humans, is an example of **maturation**.

2 Myelination

- **Myelin** is the white insulating sheath that forms around many nerve fibres, allowing increasing speed of neural communication.
- The most intense **myelination** period is directly after birth. Adolescence is also important.
- Sensory and motor areas in the brain are myelinated first. Generally, the parts of the brain that are responsible for more complex cognitive functions are myelinated afterwards.

3 Synaptogenesis and synaptic pruning

- The **synapse** is the point of communication between neurons.
- Messages are passed from axon terminals to **dendrites**.
- **Synaptogenesis** refers to the formation of new synapses between the brain's neurons, a process which occurs as the more dendrites and more branches of dendrites grow after birth.

Key Terms

- neuron
- maturation
- myelin
- synapse
- synaptogenesis
- branching of dendrites
- synaptic pruning
- frontal lobes
- prefrontal cortex

buzz words

Check Quiz

- Formation of new synapses: -----
- Tiny gap between neurons; site of communication between them: -----
- nerve cell that transmits neural impulses: -----
- fatty substance coating the nerve fibre and facilitating

Use note-taking techniques that...

5 require your brain to pose questions and find answers

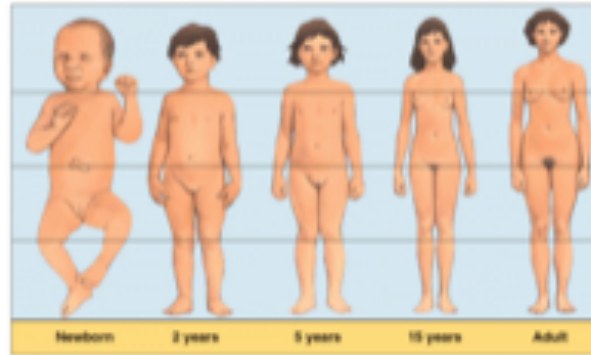
For instance, write yourself a “check quiz” during most classes.

Place it on the side or at the bottom of each page and use it for quick, structured revision.

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simple revision quiz

Check Quiz – Examples of Research Studies

survey

correlational study

case study

experiment

naturalistic observation

- a** A woman with a rare genetic illness provides details of its effects upon her.
- b** Year 11 psychology students are deprived of sleep for one night and their performance in a test is compared with that of students who were not deprived.
- c** The link between maximum daily temperature and car accidents is analysed.
- d** The flirting behaviour of young singles in a nightclub is investigated.
- e** Students fill in a questionnaire by the Department of Education to provide information about their attitudes to school.

Check Quiz – Which research method?

a A woman with a rare genetic illness provides details of its effects upon her.

case study

b Year 11 psychology students are deprived of sleep for one night and their performance in a test is compared with that of students who were not deprived.

experiment

c The link between maximum daily temperature and car accidents is analysed.

correlational
study

d The flirting behaviour of young singles in a nightclub is investigated.

naturalistic
observation

e Students fill in a questionnaire by the Department of Education to provide information about their attitudes to school.

survey

Use note-taking techniques that...

6 lead to a legible and well organised workbook, so that revision is a straightforward and even inviting task.

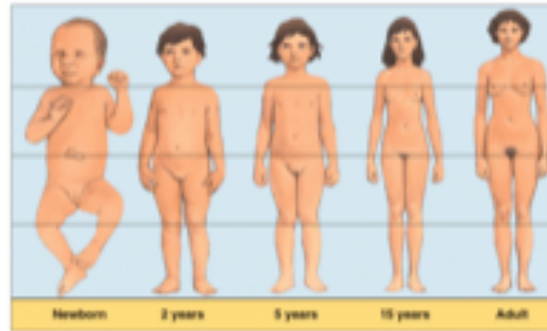
For instance, use headings, subheadings and dot points; set out your notes clearly with colour coding.

heading

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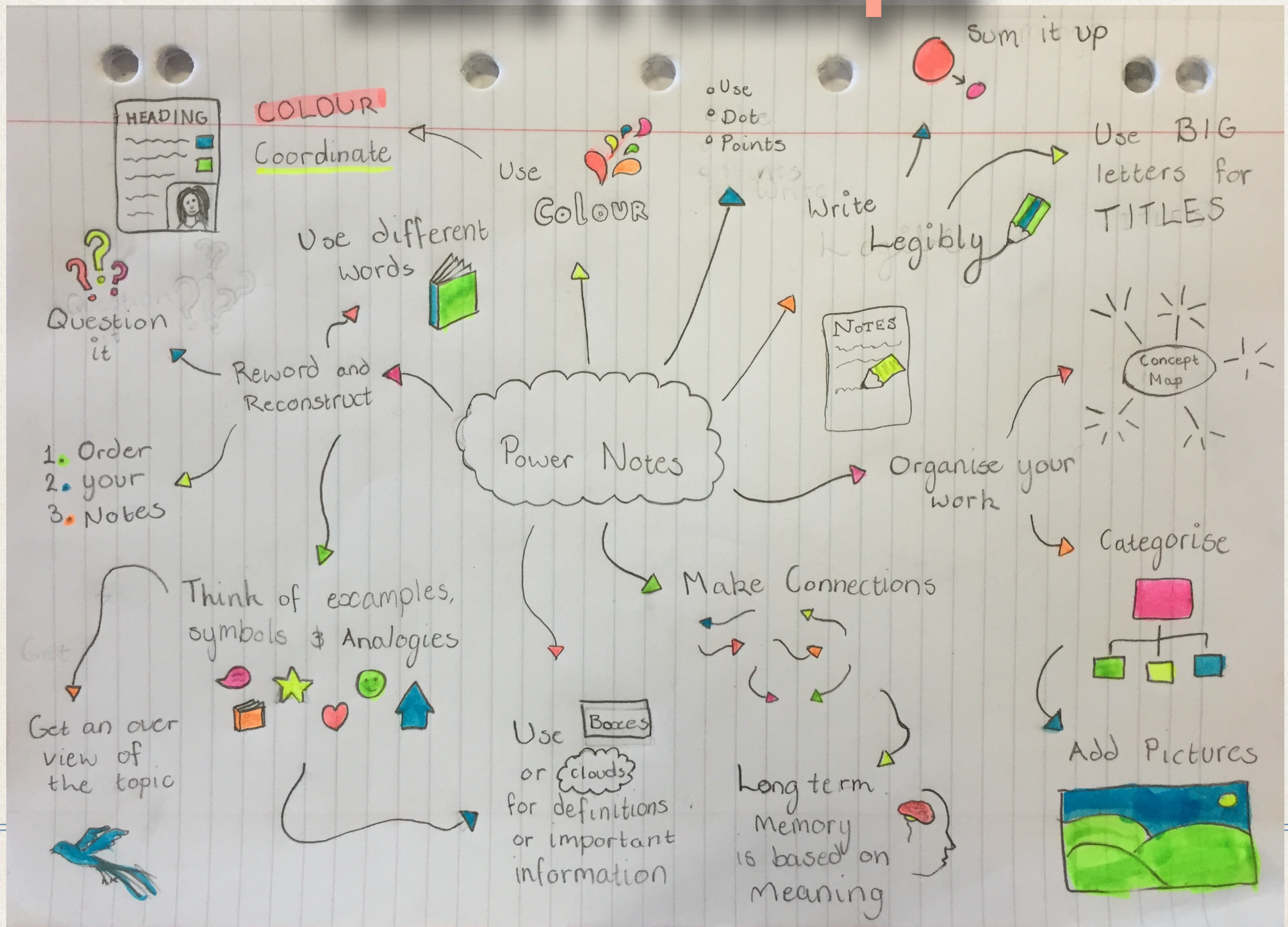
- Formation of new synapses: _____
- Tiny gap between neurons; site of communication between them: _____
- nerve cell that transmits neural impulses: _____
- fatty substance coating the nerve fibre and facilitating

buzz words

simple revision quiz

subheadings

Zahra's Example



Test your attention – Matching Quiz

- 1 Short-term memory requires... a mental states.
- 2 Long-term memory is founded on... b case study.
- 3 Feelings and thoughts are... c belief.
- 4 A synonym for thought is... d meaning.
- 5 A research method in psychology is... e attention.

Test your attention – Matching Quiz

1 A behaviour is a/n...

a affect.

2 Using synonyms, examples and analogies are all methods of...

b naturalistic observation.

3 A synonym for feeling is...

c elaboration.

4 Exploring the link between crime and temperature might require a...

d observable action.

5 Observing children at play is an example of...

e correlational study.